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- **Problem Statement ID** – SIH26149
- **Problem Statement Title**- Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization
- **Theme** - Blockchain & Cybersecurity
- **PS Category**- Software
- **Team Name** - ForenWipe

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IDEA TITLE

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• Proposed Solution

Why "Delete" Is Not Secure Erasure

Normal Deletion	Wrong-Disk Risk	NO RELIABLE VERIFICATION	FRAGMENTED FORENSIC WORKFLOW
Deletion ≠ Secure Erasure	Wrong Target → Irreversible Data Loss	Deletion alone does not prove sanitization.	Scattered evidence → Difficult to prove sanitization

PHYSICAL DEVICE AWARENESS

Model + Serial + Bus Type

Identifies the actual physical storage device before sanitization.

SANITIZATION SAFETY

Target Verification + Boot-Disk Protection

Prevents unsafe or unintended drive sanitization.

CONTROLLED WORKFLOW

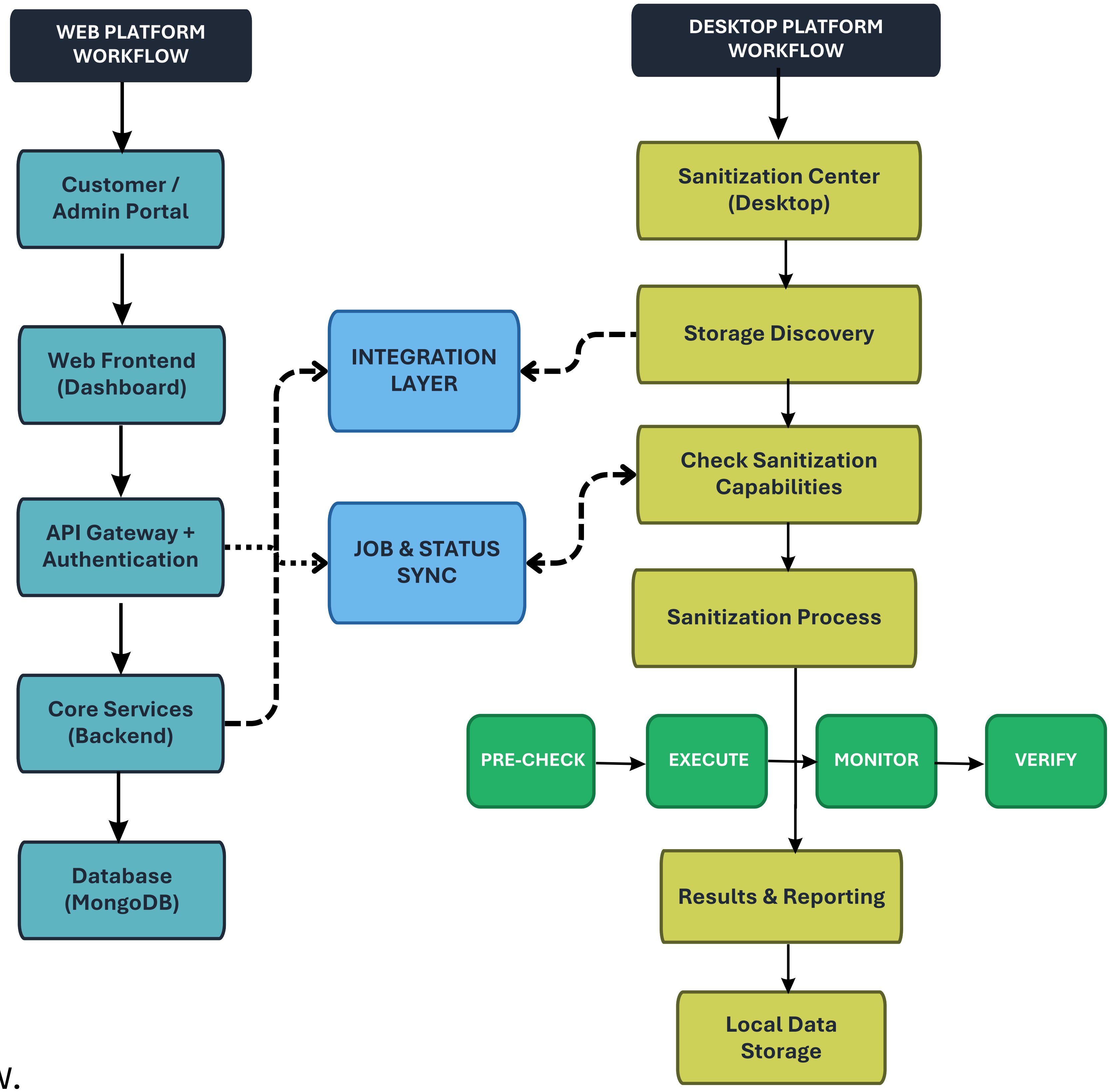
Request → Approval → Assignment → Sanitization → Verification

Connects customers, administrators and Windows workstations.

FORENSIC FILE RECOVERY

Signature-Based Forensic File Reconstruction

Recovers deleted files for forensic investigation and evidence review.



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TECHNICAL APPROACH

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1. Detect *Identify devices and gather details* → **2. Analyze** *Check capabilities and ensure safety* → **3. Securely Erase** *Device-appropriate sanitization* → **4. Verify** *Confirm successful erasure* → **5. FORENSIC RECOVERY & VALIDATION** *Recover and validate deleted-file evidence* → **6. REPORT / CERTIFY** *Generate evidence report*

1. DEVICE & HARDWARE LAYER

DETECTED STORAGE TYPES

- **HDD** *(SATA)*
- **SSD** *(SATA)*
- **NVMe** *(PCIe)*
- **USB** *(Removable)*

- Detect connected storage devices
- Identify type, model, serial number, capacity, interface
- Use Windows Storage APIs / IOCTL_STORAGE_QUERY_PROPERTY

2. SECUREWIPE CORE ENGINE

Intelligent, device-aware erasure

Device Detection & Identification
- Enumerate devices
- Read device properties
- Identify storage type

→

Safety & Capability Engine
- Analyze Device
- Verify Capability → Use only supported methods
- USB → UNKNOWN → Do not guess

→

Sanitization Engine
- HDD → Secure overwrite
- ATA/SATA → ATA commands
- NVMe → NVMe sanitize
- USB → Capability-dependent method

→

Verification
- Verify completion
- Compare expected vs actual
- Detect incomplete/failed operations

3. FORENSIC & EVIDENCE LAYER

Evidence-driven validation

Evidence Source & Context
- Select physical storage source
- Identify source & storage type
- Preserve recovery context

→

Raw Storage Scanning
- Process storage in 4 MB chunks
- Preserve boundary bytes between chunks
- Scan raw data for file signatures

→

Signature-Based JPEG Carving
- Detect JPEG signature
- Extract candidate artifact ranges
- Recover files across chunk boundaries

→

Multi-Layer Recovery Validation
- Header / Footer validation
- Size consistency check
- JPEG structural validation
- Decoder / decodability check

→

Evidence Integrity & Confidence
- Generate SHA-256 artifact hash
- 0–100 rule-base confidence score
- Generate explainable confidence reasons

4. APPLICATION & REPORTING LAYER

User interface, audit and compliance

Qt Frontend (GUI)
- Device overview
- Configure & start wipe
- View results

→

Backend / Core Engine
- Execute operations
- Manage logs & data
- Interface with Windows APIs
- Handle verification & forensics

→

Audit Logs
- Store complete wipe record
- Device identity, method, timestamps
- Verification & forensic results
- Cryptographic integrity (hashes)

→

Certification
- Generate verifiable report
- Tamper-evident records
- Compliance-ready output

FEASIBILITY AND VIABILITY

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FEASIBILITY

DEVICE INTELLIGENCE

Identifies connected storage devices and determines the appropriate workflow before processing.

Real-device assessment

SAFE SANITIZATION

Applies controlled, device-aware sanitization while preventing unsafe operations on the wrong device.

Safe • Controlled • Adaptive

DEEP FORENSIC RECOVERY

Examines raw storage to recover deleted file remnants, including data that crosses normal scan boundaries.

Raw Storage • File Carving

EVIDENCE VALIDATION

Validates recovered files through multiple checks to distinguish complete, usable evidence from corrupted remnants.

Structure • Decodability

REAL-WORLD VIABILITY

TRUSTED DISPOSAL

Sanitization provides confidence before a device is reused, sold or disposed.

RECOVERY ASSURANCE

Forensic examination helps determine whether recognizable data remains recoverable from storage media.

EVIDENCE-BASED DECISIONS

Recovery findings help organizations assess residual data exposure before reuse or disposal.

TRACEABLE WORKFLOW

Processing results and forensic findings can be maintained as structured records for review.

BUSINESS MARKET POTENTIAL

SECURE DISPOSAL & ITAD

Secure sanitization + forensic verification for organizations, refurbishers & ITAD.

Bulk • Subscription • Service

FORENSIC & INCIDENT SERVICES

Deleted-data recovery + evidence validation for investigations.

Recovery • Evidence • Service

ENTERPRISE PLATFORM

Integrated sanitization + forensic assessment for enterprise workflows.

SaaS • Enterprise License • API

Technical Validation

SANITIZATION | Real Windows Hardware • SATA + USB Testing • Capability Safety

FORENSICS | Real Storage Data Recovery • File Carving • Recovery Validation

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IMPACT AND BENEFITS

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Transforming storage sanitization and forensic recovery into a safe, intelligent and auditable workflow.

01 DATA PRIVACY

- Reduces risk of sensitive data exposure
- Supports responsible disposal of retired devices
- Protects organizational, government & personal information

02 SAFER DISPOSAL

- Reduces risk during device retirement
- Helps prevent accidental data loss
- Makes sanitization easier to perform correctly

03 OPERATIONAL EFFICIENCY

- Reduces manual inspection effort
- Brings sanitization into one workflow
- Helps standardize repeatable operations

04 COMPLIANCE & ACCOUNTABILITY

- Creates a record of sanitization activity
- Supports internal audits and organizational policies
- Makes disposal decisions easier to review

05 DIGITAL FORENSICS

- Supports investigation of deleted data
- Helps analysts review recovered evidence
- Enables evidence-oriented recovery workflows
- Useful for incident response and investigations

Safer Disposal | Standardized Operations | Accountable Handling | Forensic Support

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RESEARCH AND REFERENCES

NIST SP 800-88 Rev. 2 — Media Sanitization

Sanitization principles, assurance and verification guidance.
Applied to: sanitization workflow & verification design.

Guidelines for Media Sanitization

Microsoft Windows Storage APIs

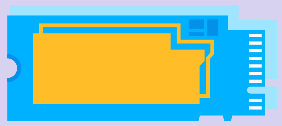
IOCTL_STORAGE_QUERY_PROPERTY + STORAGE_DEVICE_DESCRIPTOR.
Applied to: model, serial, bus type & physical-device discovery.

STORAGE_DEVICE_DESCRIPTOR structure (winioctl.h)

SCSI / ATA Pass-Through Research

Low-level command paths for storage-device interrogation and operations.
Research basis: Low-level storage command interrogation & device-specific operations.

SCSI_PASS_THROUGH structure (ntddscsi.h)



NVMe Command & Sanitization Research

Identify Controller, SANICAP and NVMe administrative commands.
Applied to: NVMe capability-aware sanitization design.

Working with NVMe drives

Digital Forensics & File Carving

The Sleuth Kit / PhotoRec research on signature-based recovery and raw-data analysis.
Research focus: Signature-based carving and raw-data analysis.
Applied to: JPEG carving, validation, SHA-256 hashing & confidence scoring.

Sleuth Kit Labs (open source project)